

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1838

Jack McDonough 138 kV
September 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1838. The Interconnection Customer proposes a 500.0 MW battery energy storage facility interconnecting at the Jack McDonough 138 kV substation in the Maryland area. The request was received on July 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 420 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 27 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$74,232,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Trumann Substation – Rossmoor S 138kV
Contingency: Tiger Bay – Air Force Base Substation 345kV
Loading: 158.1% (Rating: 245 MVA)
DFAX: 5.4%
Pre-project loading: 147.1%
2. Facility: Low Gap – Flint River 138kV line
Contingency: Low Gap – Canovanas Pueblo 2401 138kV line
Loading: 154.4% (Rating: 182 MVA)
DFAX: 29.4%
Pre-project loading: 73.5%
3. Facility: Tramway – Rossmoor South 138kV
Contingency: Tiger Bay – Air Force Base 345kV
Loading: 142.7% (Rating: 313 MVA)
DFAX: 5.4%
Pre-project loading: 134.2%
4. Facility: Bondurant – Tramway 138kV L
Contingency: Bondurant – Tramway 138kV Cir. 1
Loading: 135.1% (Rating: 376 MVA)
DFAX: 7.5%
Pre-project loading: 125.1%
5. Facility: Lamont – Dundee 138kV
Contingency: Peachland – Shepherdstown 138kV
Loading: 133.3% (Rating: 215 MVA)
DFAX: 16.9%
Pre-project loading: 93.9%

6. Facility: Bondurant – Montgomery Village Substation 138kV Circuit 2
Contingency: Montgomery Village – Bondurant 138kV Line 1
Loading: 133.3% (Rating: 396 MVA)
DFAX: 7.5%
Pre-project loading: 123.8%
7. Facility: Cheatham – Canovanas Pueblo 2401 138kV
Contingency: Artesia Country Club Substation – Air Force Base 345kV
Loading: 132.9% (Rating: 305 MVA)
DFAX: 31.8%
Pre-project loading: 80.7%
8. Facility: Tramway – Bondurant 138kV Line 1
Contingency: Bondurant Substation – Tramway 138kV Line 2
Loading: 130.1% (Rating: 391 MVA)
DFAX: 7.5%
Pre-project loading: 120.5%
9. Facility: Jack McDonough – Lamont 138kV L
Contingency: Artesia Country Club – Air Force Base 345kV
Loading: 126.2% (Rating: 556 MVA)
DFAX: 51.2%
Pre-project loading: 80.1%
10. Facility: Low Gap – Canovanas Pueblo 2401 138kV
Contingency: Air Force Base Substation – Artesia Country Club 345kV
Loading: 125.9% (Rating: 358 MVA)
DFAX: 31.8%
Pre-project loading: 81.4%
11. Facility: Lcec No. 5 – Opequon 138kV
Contingency: Clark County – Kearneyville 345kV
Loading: 125.3% (Rating: 515 MVA)
DFAX: 5.2%
Pre-project loading: 120.3%
12. Facility: North Mesa – Opequon 138kV Line
Contingency: Clark County Substation – Kearneyville 345kV line
Loading: 124.4% (Rating: 529 MVA)
DFAX: 5.2%
Pre-project loading: 119.5%
13. Facility: Bondurant – Montgomery Village 138kV
Contingency: Montgomery Village – Bondurant 138kV
Loading: 122.0% (Rating: 433 MVA)
DFAX: 7.5%
Pre-project loading: 113.3%
14. Facility: Scarcello – Rokville 138kV
Contingency: Tramway – Rossmoor South 138kV
Loading: 117.3% (Rating: 230 MVA)
DFAX: 5.8%
Pre-project loading: 104.6%
15. Facility: Jack McDonough – Low Gap 138kV
Contingency: Air Force Base – Artesia Country Club 345kV
Loading: 113.5% (Rating: 581 MVA)
DFAX: 48.8%
Pre-project loading: 71.6%
16. Facility: Tramway – Scarcello 138kV
Contingency: Rossmoor S – Tramway 138kV

Loading: 112.6% (Rating: 306 MVA)

DFAX: 5.8%

Pre-project loading: 103.0%

17. Facility: Lamont – University City 138kV line

Contingency: Clark County – Kearneyville 345kV

Loading: 112.1% (Rating: 467 MVA)

DFAX: 11.1%

Pre-project loading: 100.3%

18. Facility: University City – Richford 138kV L

Contingency: Clark County – Kearneyville 345kV

Loading: 110.6% (Rating: 508 MVA)

DFAX: 11.1%

Pre-project loading: 99.7%

19. Facility: Richford – Sun City 138kV Line

Contingency: Clark County – Kearneyville 345kV

Loading: 110.1% (Rating: 525 MVA)

DFAX: 11.1%

Pre-project loading: 99.5%

20. Facility: Peachland – Sun City 138kV L

Contingency: Clark County – Kearneyville 345kV

Loading: 109.8% (Rating: 535 MVA)

DFAX: 11.1%

Pre-project loading: 99.4%

21. Facility: Peacland – Shepherdstown 138kV line

Contingency: Clark County – Kearneyville 345kV

Loading: 109.5% (Rating: 544 MVA)

DFAX: 11.1%

Pre-project loading: 99.3%

22. Facility: International Paper Valliant Ok – Kearneyville 345kV

Contingency: St. John – Rio Dell 138kV L

Loading: 108.2% (Rating: 2,128 MVA)

DFAX: 10.6%

Pre-project loading: 105.7%

23. Facility: Rockville – Phillips Pump 1 138kV

Contingency: Rossmoor S – Tramway 138kV

Loading: 108.2% (Rating: 225 MVA)

DFAX: 5.8%

Pre-project loading: 95.2%

24. Facility: Carlotta – West Barlett 138kV

Contingency: Lamont – Jack McDonough 138kV Line

Loading: 107.8% (Rating: 239 MVA)

DFAX: 28.4%

Pre-project loading: 48.3%

25. Facility: W Barlett – Cheatham 138kV

Contingency: Jack McDonough – Lamont 138kV Line

Loading: 105.8% (Rating: 236 MVA)

DFAX: 28.4%

Pre-project loading: 45.5%

26. Facility: Clark County – Kearneyville 345kV L

Contingency: Shepherdstown – Peachland 138kV

Loading: 105.0% (Rating: 2,021 MVA)

DFAX: 12.0%

Pre-project loading: 102.0%

27. Facility: Skykomish – Athalmer Tap 138kV Line

Contingency: Pineville – Innerarity 138kV

Loading: 100.6% (Rating: 254 MVA)

DFAX: 6.4%

Pre-project loading: 88.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Trumann Substation – Rossmoor S 138kV	Rebuild line with higher-capacity conductor	\$6,619,000	60% (\$3,946,000)
Low Gap – Flint River 138kV line	Rebuild line with higher-capacity conductor	\$10,606,000	100% (\$10,606,000)
Tramway – Rossmoor South 138kV	Rebuild line with higher-capacity conductor	\$2,383,000	60% (\$1,420,000)
Tramway – Bondurant 138kV Line 2	Rebuild line with higher-capacity conductor	\$1,172,000	72% (\$846,000)
Lamont – Dundee 138kV	Rebuild line with higher-capacity conductor	\$5,427,000	100% (\$5,427,000)
Montgomery Village – Bondurant 138kV	Rebuild line with higher-capacity conductor	\$7,260,000	72% (\$5,240,000)
Cheatham – Canovanas Pueblo 2401 138kV	Rebuild line with higher-capacity conductor	\$2,973,000	96% (\$2,863,000)
Tramway – Bondurant 138kV #1	Rebuild line with higher-capacity conductor	\$2,332,000	72% (\$1,683,000)
Jack McDonough – Lamont 138kV L	Rebuild line with higher-capacity conductor	\$6,450,000	78% (\$5,012,000)
Low Gap Substation – Canovanaspueblo 2401 138kV	Rebuild line with higher-capacity conductor	\$6,202,000	96% (\$5,970,000)
Lcec No. 5 – Opequon 138kV	Rebuild line with higher-capacity conductor	\$3,485,000	9% (\$316,000)
North Mesa – Opequon 138kV Line	Rebuild line with higher-capacity conductor	\$3,300,000	9% (\$299,000)
Bondurant – Montgomery Village 138kV	Rebuild line with higher-capacity conductor	\$8,869,000	72% (\$6,401,000)
Scarcello – Rokville 138kV	Rebuild line with higher-capacity conductor	\$913,000	100% (\$913,000)
Jak McDonough – Low Gap 138kV	Rebuild line with higher-capacity conductor	\$1,612,000	77% (\$1,238,000)
Tramway Substation – Scarcello 138kV line	Rebuild line with higher-capacity conductor	\$1,294,000	100% (\$1,294,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lamont – University City 138kV line	Rebuild line with higher-capacity conductor	\$19,177,000	34% (\$6,420,000)
Richford – University City 138kV	Rebuild line with higher-capacity conductor	\$5,783,000	34% (\$1,936,000)
Richford – Sun City 138kV Line	Rebuild line with higher-capacity conductor	\$8,782,000	34% (\$2,940,000)
Peachland – Sun City 138kV L	Reconductor line	\$1,228,000	34% (\$411,000)
Peacloud – Shepherdstown 138kV line	Reconductor line	\$747,000	34% (\$250,000)
International Paper Valliant Ok – Kearneyville 345kV	Reconductor line	\$19,450,000	5% (\$962,000)
Rockville – Phillips Pump 1 Substation 138kV	Reconductor line	\$692,000	100% (\$692,000)
Carlotta – West Barlett 138kV	Reconductor line	\$2,286,000	100% (\$2,286,000)
W Barlett – Cheatham 138kV	Reconductor line	\$1,261,000	100% (\$1,261,000)
Clark County – Kearneyville 345kV L	Reconductor line	\$15,431,000	6% (\$872,000)
Skykomish – Athalmer Tap 138kV Line	Reconductor line	\$2,728,000	100% (\$2,728,000)
	Total allocated to project		\$74,232,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.